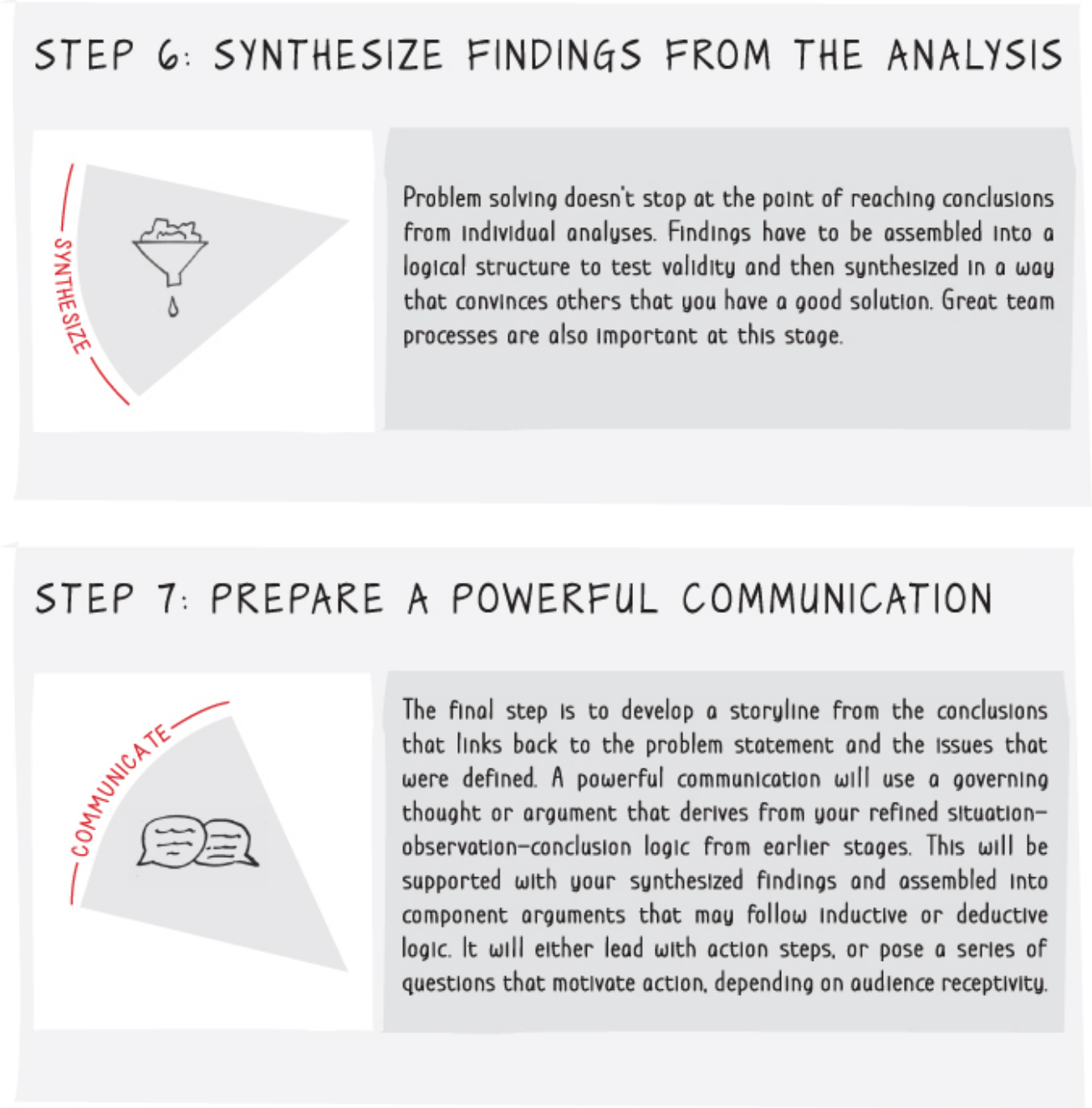


EXHIBIT 1.1 The bulletproof problem solving cycle

Prepare for an Avalanche of Trees!

We use logic or issue trees to visualize and disaggregate problems. We employ several types, including hypothesis trees and decision trees, as you will see in the cases we present throughout this book. We

learned the power of logic trees at McKinsey and continue to find them essential to good problem solving. Why? Because they do the following:

Provide a clear visual representation of the problem so that everyone can understand the component parts.

Done correctly, they are holistic in the sense that everything relevant is captured in the tree.

Lead to clear hypotheses that can be tested with data and analysis.

Our logic trees are sometimes simple and sometimes highly complex. But they all started on a sketchpad or a whiteboard.

Let's Start with Some Case Studies

To illustrate the bulletproof problem solving process, we chose some case studies that represent classes of problems that many of our readers will face, and that exhibit the power and utility of the process described in detail over the next several chapters.

> Is Sydney airport capacity adequate for the future?
>
> Should I install solar panels on my roof now?
>
> Where should I move?
>
> Should a start-up raise its prices?
>
> Should I support a K–12 school education levy in my town?

These relatively simple cases will outline each of the seven problem solving steps, but with a focus on the use of logic trees to help you represent the problem and break it into manageable parts. Later chapters will go into the fine points of the other steps in more detail and for more complicated problems.

Case 1: Does Sydney Airport Have Adequate Capacity?

When Rob was the lead partner in recruiting for the Australian and New Zealand practice of McKinsey, the consulting firm made the decision to look beyond traditional hires with MBAs to try to attract clever physicists, scientists, lawyers, engineers, and liberal arts graduates. Discussing business cases in interviews put many of the potential hires at a disadvantage. So his recruiting team came up with a non-business case that they called the Sydney Airport case. It is pretty simple, but it is a good way to show the seven-steps method.

All of the candidates had flown into Sydney Airport and were aware of discussions in the newspapers about whether another airport was needed at that time. Sydney Airport has two of the 10 busiest air routes in the world, so this is a real-world example. At the interviews the candidates were given a simple problem definition (step 1 problem definition): “Will Sydney Airport capacity be adequate in the future?” and asked how they would think about that question. The problem statement was bounded around

passenger airport capacity, so the candidates didn't have to spend a lot of time on policy factors that might warrant a second airport, such as greater accessibility, safety, or environmental factors like noise, or even alternatives like a very fast train link between major cities. As we'll see later, the boundaries on problem definition are really important to agree on up front.

Candidates would often ask a clarifying question or two and then outline their approach to addressing the issue. So what was Rob's team looking for? They wanted to see if the candidates used a logical structure to help them solve the problem. It's much easier to show the parts of the problem in written form, so we encouraged candidates to use a whiteboard or pad of paper. It is usually a trial and error process to get the breakdown right to solve the problem. This is step 2, problem disaggregation, and Exhibit 1.2 shows a simple first cut.



EXHIBIT 1.2

In this case, the simplest possible way to cleave the problem is to define airport capacity as supply (of landing slots) less demand. You could have a more complicated tree with competition from other ways to

get to Sydney (and you might get extra credit for showing how those affect demand), but it probably isn't necessary in this relatively simple case.

A good candidate would dig a little deeper of course. Exhibit 1.3 shows one way of defining airport supply capacity (number of runways, capacity of each runway, and utilization) and demand (Sydney's share of regional demand). In the short term, the number of runways is fixed, and so is runway capacity (defined mostly by aircraft type).



EXHIBIT 1.3

Candidates would typically explain their approach to modeling demand growth by making different assumptions about gross domestic product (GDP) growth, fuel costs, and relative location attractiveness

of Sydney relative to other destinations (see Exhibit 1.4).



EXHIBIT 1.4

But the most productive approach to this problem is to go deeper into runway utilization, as it is one of the few variables that can be actively managed by transportation planners. Runway utilization is determined by hours of operation, spacing between aircraft movements, and the number of people per airplane. Hours of operation are limited by curfew periods, weather, and maintenance. Thinking about how you could vary these is the heart of steps 4 (workplanning) and 5 (analysis).

The answers Rob most liked were the ones where candidates would say something along these lines:

> *Runway utilization is the key so I'd be looking at operating hours, planes per hour, and the people per plane. You probably can't do much with operating hours because there are curfew restrictions between midnight and 6 a.m. because of the residents nearby. With planes per hour—a core variable for utilization—I'd want to see if they could safely reduce further the time between takeoffs and landings. The third factor is the people per plane and that comes down to slot pricing favoring larger planes and policy about light aircraft use at peak hours (steps 6 and 7 synthesis and storytelling)*